

CRUD application for employee details

1. Software installations required

1.1 Front-end

- **Node.js:** A back-end JavaScript runtime environment, that lets us executes JavaScript code outside a web browser.
- **Visual studio code:** A source-code editor which supports debugging, syntax highlighting, intelligent code completion, snippets, code refactoring, and embedded Git.

1.2 Back-end

- **Java Development kit:** A development environment for building applications, applets, and components using the Java programming language. JDK includes tools useful for developing and testing programs written in the Java programming language.
- **Spring tool suite:** An IDE with Eclipse-based development environment to develop Spring applications. It provides a ready-to-use environment to implement, run, deploy, and debug the application.
- **MySQL-router, server, workbench:** MySQL is a relational database management system, based on the Structured Query Language. Workbench is a visual database design tool that integrates SQL development, administration, database design, creation and maintenance into a single integrated development environment for the MySQL database system.
- **Postman:** A software testing API (Application Programming Interface) platform to build, test, design, modify, and document APIs. It is a simple Graphic User Interface for sending and viewing HTTP requests and responses.

1.3 Git and github

Git is a popular version control system. It is used for tracking code changes, tracking who made changes and coding collaboration. It lets us manage projects with repositories, clone a project to work on a local copy, control and track changes with staging and committing, branch and merge to allow for work on different parts and versions of a project, pull the latest version of the project to a local copy and push local updates to the main project.

GitHub, Inc. is a platform and cloud-based service for software development and version control using Git, allowing developers to store and manage their code. It provides the distributed version control of Git plus access control, bug tracking, software feature requests, task management, continuous integration, and wikis for every project.

2. Technologies used

2.1 Front-end

- **React JS:** A open-source front-end JavaScript library for building user interfaces based on components.
- **HTML and CSS:** These are used to display and style the contents of the web page or web application.

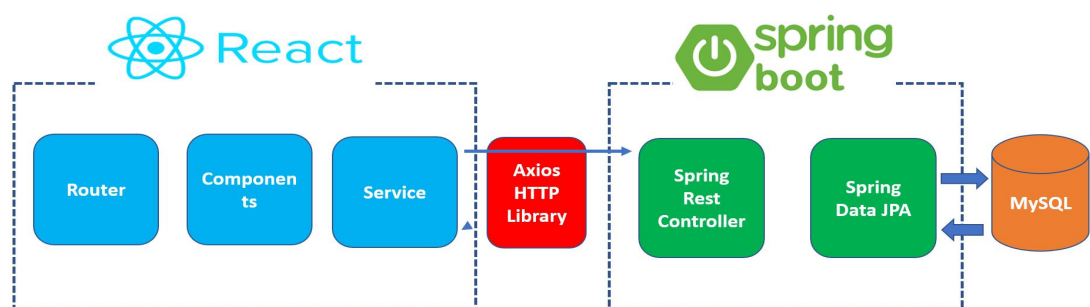
2.2 Backed

- **Java Spring boot:** Java spring framework is a open source, enterprise level framework for creating standalone, production grade applications that run on java virtual machine. Java spring boot is a tool that makes developing web application and micro services with spring framework faster and easier.
- **SQL:** SQL is a standard language for storing, manipulating and retrieving data in databases. It stands for Structured Query Language. RDBMS is the basis for SQL and the data in RDBMS is stored in database objects called tables. A table is a collection of related data entries and it consists of columns and rows.

3. Working

Working of the crud application follows the below diagram.

ReactJS + Spring Boot CRUD Full Stack

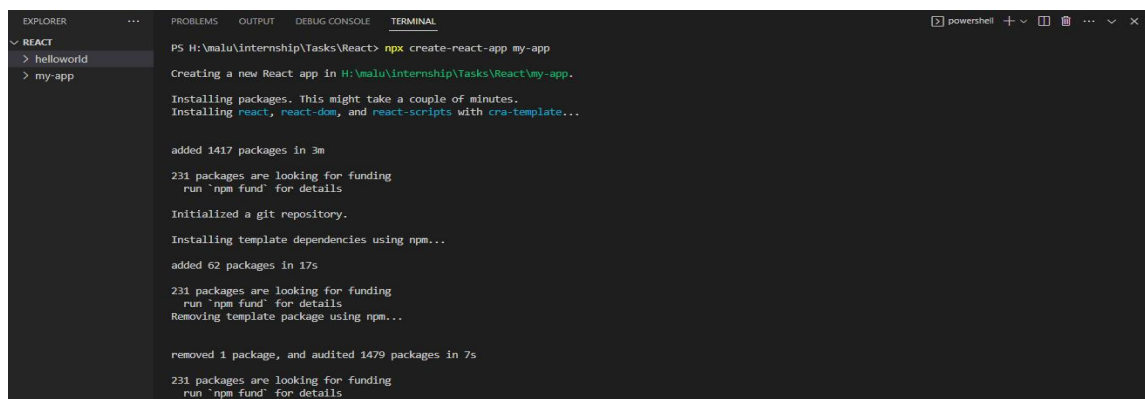


4. Steps involved

4.1 Creating a database with employee details in MySQL workbench and creating a react page to list the employee details.

a) Creating the react-app

- To create a react app the latest versions of node and npm must be installed into the system. Then in the terminal of the visual studio code we can use the command **'npx create-react-app frontend'** to create a sample react app named frontend with its existing libraries and folders in the frontend directory.
- We can make the required changes to the existing application by making changes in the App.js file in the directory. After this we can use the command **'npm start'** in the frontend directory to run the react application.



```
PS H:\malu\Internship\Tasks\React> npx create-react-app my-app
Creating a new React app in H:\malu\Internship\Tasks\React\my-app.
Installing packages. This might take a couple of minutes.
Installing react, react-dom, and react-scripts with cra-template...

added 1417 packages in 3m

231 packages are looking for funding
  run 'npm fund' for details

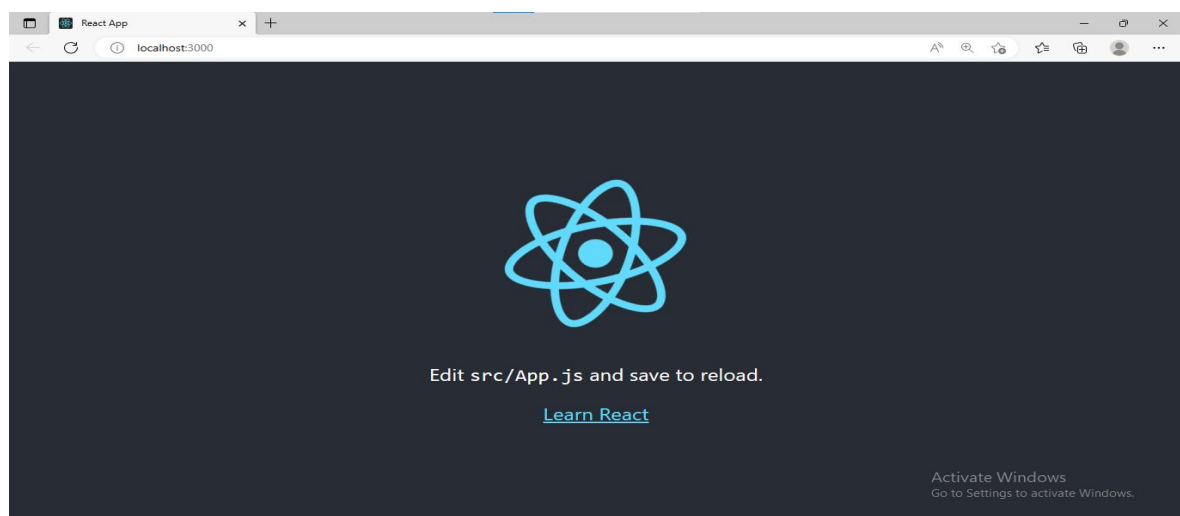
Initialized a git repository.
Installing template dependencies using npm...
added 62 packages in 17s

231 packages are looking for funding
  run 'npm fund' for details
Removing template package using npm...

removed 1 package, and audited 1479 packages in 7s

231 packages are looking for funding
  run 'npm fund' for details
```

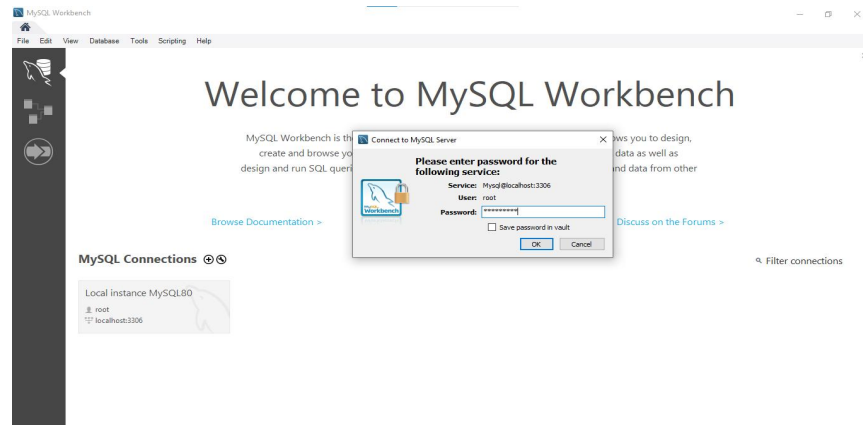
1.1 An example of creating a react app called my-app



1.2 Sample react app on running npm start in the project directory

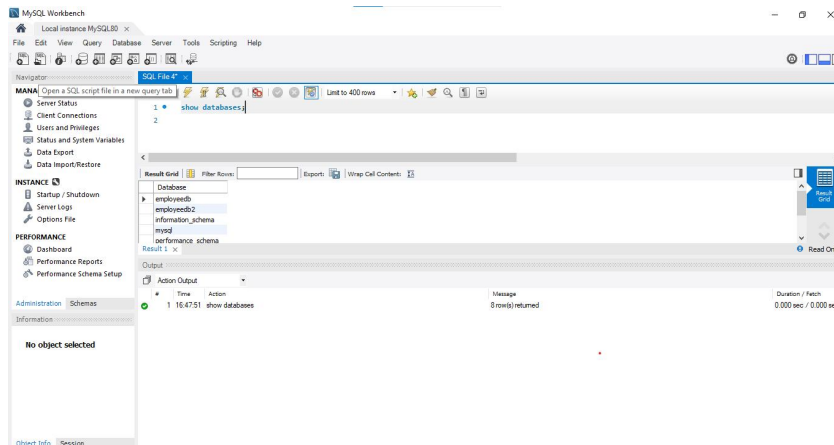
b) Creating a database manually in MySQL workbench

- We first launch your MySQL workbench on our system and click on the local instance and enter the password. Then we create a new tab to execute sql queries. We first check for the existing data bases using the query, **show databases**. Then



1.3 On launching MySQL workbench

we either choose from the existing databases or create a new database using the query, **create database employeeedb** which creates a database of the name employeeedb.

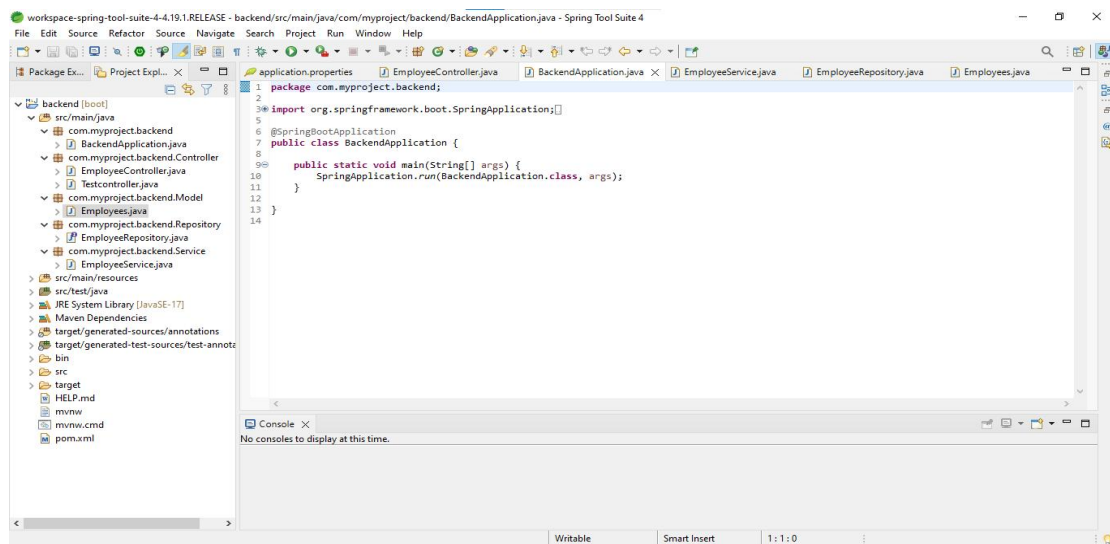


1.4 Executing queries

- Now we create a new table using the query, **create table employees(emp_id integer, name varchar(30), role varchar(30), location varchar(30))**. We can add new employee details using the query, insert into employee values(1, “Rachel williams”, “Senior developer”, “Bangalore”). Similarly we add other employee names into the table.

c) Creating a spring boot application

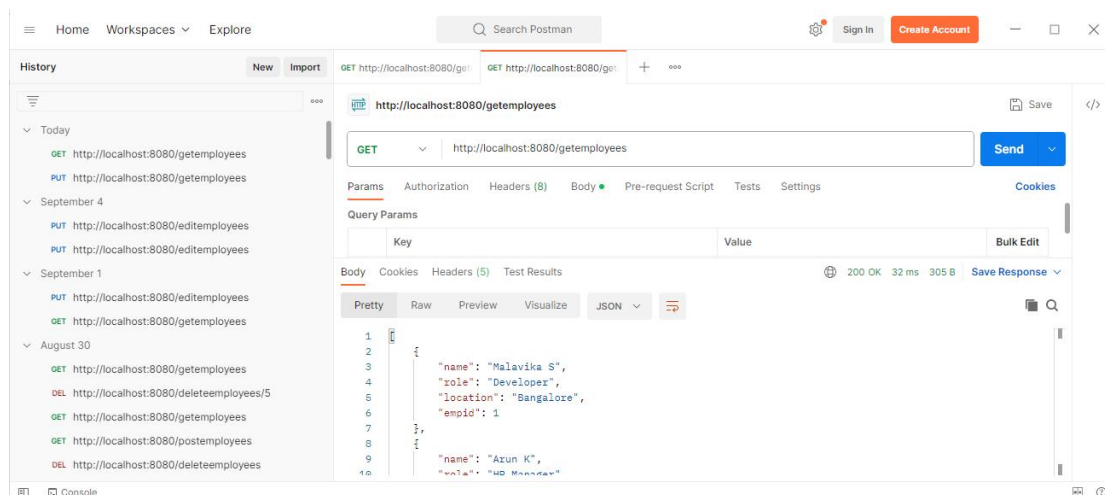
- We must launch the spring suite tool and create a new spring starter project and select required properties. A project is created with all the folders with the specified properties and dependencies. Under the order of folders src/main/java we find the main application.java file which contains the main function.
- We now create different packages in our project to maintain a proper structure. First we create the model class called **Employees** which will contain different data objects with respect to the table we have created in the database.
- Similarly next we create a service class called **EmployeeService** which are important parts of a Spring Boot application and they hold business logic. Then we add the repository interface called the **EmployeeRepository** which provides data access abstraction and lets us access to the JPA repository which contains API of CrudRepository and PagingAndSortingRepository. So it contains API for basic CRUD operations and also API for pagination and sorting.
- Now finally we create the controller class called **EmployeeController** which is responsible for handling incoming requests and redirecting them to a suitable service.



1.5 Structuring the project

d) Displaying the employee table in the react page

- We first create an **GetMapping** annotation in the controller class which creates an **/getemployees** API. We can test to see if the API is working using the Postman software by giving **http://localhost:8080/getemployees**.



1.6 In postman

- This API is accessed through the react application using the fetch request and the **useEffect** hook to display the data in the table format by using the map function.



1.7 Displaying details from the database

4.2 Creating new employee with an employee form and reflecting it in the database.

- Creating a form in react and connecting to back end through creation of API**
 - We first create a create new employee button and onClick we open a popup form as shown in the image to take new employee details. We use **useState** to change the field states everytime values are entered.

The screenshot shows a web browser window with the title 'Employee Details App' and the URL 'localhost:3000'. A button labeled 'Create new employee' is in the top left. A modal form titled 'New employee details' is centered. It contains the following fields:

- Emp. ID:
- Name:
- Role:
- Location:

An 'Add employee' button is located at the bottom of the form.

1.8 Form for entering employee details

- In the backend springboot application we use the **PostMapping** annotation to create an new API to post data into the database. Similarly we use the axios library in the react applications to post the entered data using the API created. Axios is mainly used to send asynchronous HTTP requests to REST endpoints.

b) Adding validations to the input form using react

- We add validations to ensure that user input is clean, correct, and useful. Typical validation tasks to check if the user has filled in all required fields, if the user has filled in the valid input etc.

The screenshot shows the same web application, but now with validation errors. The 'New employee details' form is still open, but the input values are different:

- Emp. ID:
- Name:
- Role:
- Location:

Red error messages are displayed below the Name, Role, and Location fields:

- Below Name: Name should not contain numbers or special characters!
- Below Role: Role is required!
- Below Location: Location is required!

The 'Add employee' button remains at the bottom.

1.9 Form validations

- We can use regular expressions for validating specific fields in the user input form. A Regular Expression is an object that describes a pattern of characters. Regular Expressions are used to perform pattern-matching and search-and-replace functions on text. So, in this area JavaScript plays a major role in validating the values.

4.3 Adding delete and edit functionality on existing employee details.

a) Adding delete and edit buttons using react

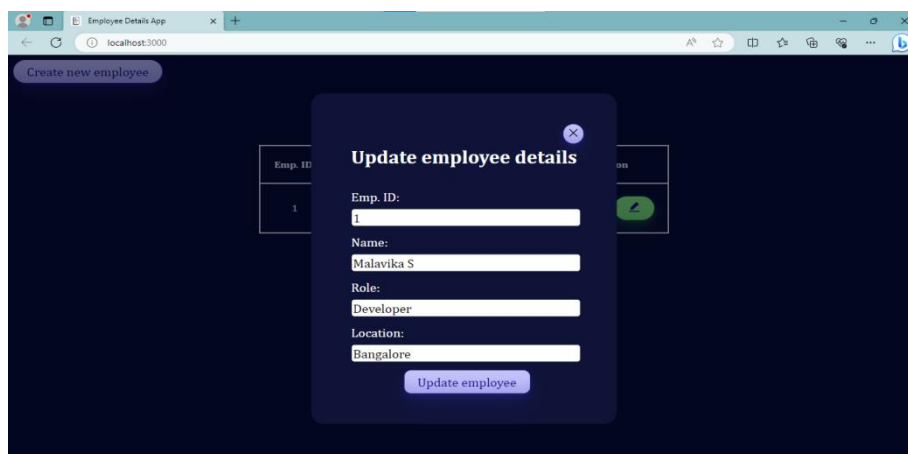
We can use the react icons to facilitate our delete and edit buttons. React-icons is a small library that helps us add icons (from all different icon libraries) to our React apps. It delivers the icons to our app as components so they're easier to work with, and it lets us style them so they're consistent with the overall style of our app.

Emp. ID.	Name	Role	Location	Action
1	Malavika S	Developer	Bangalore	 

1.10 Edit and delete icons using react

b) Adding the edit form and reflecting changes both in front end and back end.

We can reuse the the same form code as that of the adding new users and also use the same API that we created to post new employees into the table. We create a form where the existing data is already filled into the fields and can be editing the new values . We can incorporate this using the **useState** hook by setting the state of the fields to existing data everytime the edit button is clicked.

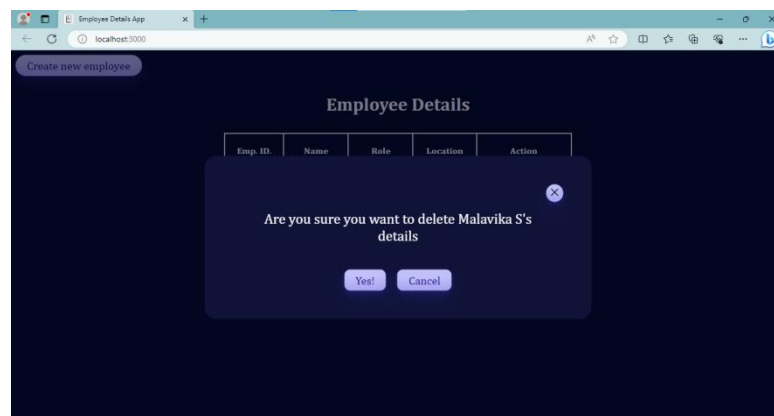


The screenshot displays a web browser window with the title 'Employee Details App'. The address bar shows 'localhost:3000'. The main content area features a dark blue background. On the left, there is a button labeled 'Create new employee'. In the center, a modal titled 'Update employee details' is open, containing a close button (X) in the top right corner. The modal has four input fields: 'Emp. ID' with the value '1', 'Name' with 'Malavika S', 'Role' with 'Developer', and 'Location' with 'Bangalore'. Below these fields is a button labeled 'Update employee'. To the right of the modal, a portion of a table is visible, showing columns for 'Emp. ID', 'Name', 'Role', and 'Location', with the first row containing the values '1', 'Malavika S', 'Developer', and 'Bangalore'.

1.11 Editing form

c) **Deleting employees with confirmation message and reflecting changes in both front end and back end.**

When we delete an employee, we first ask for confirmation using a pop up to make sure the user really wants make such a change to the database. Here we use the **DeleteMapping** annotation in the springboot application to create an **/deleteemployees** API and use an axios delete request to delete the employee details.



1.12 Confirmation message to delete the employee details

4.4 Pushing Project repository into github

Pushing code to GitHub means to upload our project code to the GitHub.com code-hosting service. To successfully push code to GitHub, we will need to first create a remote repository. We can do this directly in our GitHub account. The URL available in the repository can be used to push our code using different commands in our terminal. Some of the benefits of github is that it makes it easy to contribute to our open-source projects, it lets us showcase our work, track changes in your code across versions and also GitHub can integrate with common platforms such as Amazon and Google Cloud, services such as Code Climate to track our feedback, and can highlight syntax in over 200 different programming languages.